

Use of data augmentation in household transmission models to account for missing data

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Abstract

Epidemiological studies employing household-based recruitment are often used to estimate transmission of pathogens between household members and from the community. These transmission models can generate key insights into pathogen spread and can inform the effectiveness of interventions targeted at households. However, incomplete enrollment of household members due to survey design or non-participation leads to missing data that can bias estimates of household transmission. In this study, we extend a final-size household model using serological (antibody) status of the study population to infer within-household and community infection. We introduce a data augmentation step in a Bayesian framework to account for the serological status of missing household members and their effect on household transmission. We apply this model to a household-based serosurvey of chikungunya virus conducted in a cohort study in Ponce, Puerto Rico. In this study, 1,229 out of 3,853 individuals were enrolled as part of a complete household (509/2,171 households). The probability of transmission from an infectious to susceptible household member was estimated to be 0.13 (95% CrI 0.11–0.15), compared to 0.16 (95% CrI 0.12–0.20) using complete households only. Use of screens and air conditioning was protective against infection from outside the household, but there was no evidence that these factors were associated with within-household transmission. Spatial dependence in seropositivity between members of different households extended to XXm. This framework can be readily applied to household-based final-size models to recover mechanistic parameters for use in transmission models to inform intervention policy and deployment.

Keywords: Bayesian inference, data augmentation, missing data, household serology, chikungunya

1 Introduction

Chikungunya virus (CHIKV) is a mosquito-borne alphavirus transmitted primarily by *Aedes aegypti* and *Aedes albopictus*, and the cause of an acute febrile illness whose hallmark is severe and frequently persistent polyarthralgia [1]. First isolated in East Africa in the early 1950s, the virus circulated in Africa and Asia for decades before a series of adaptive mutations expanded its vector range and reshaped its epidemiology [2, 3]. Its arrival in the Caribbean in late 2013 initiated explosive transmission throughout the Americas, where a largely immunologically naive population sustained outbreaks across the Caribbean, Central, and South America within two years [4, 5, 6]. Since this introduction, dengue, chikungunya, and Zika viruses have circulated across the same regions and the same vector, yet have displayed markedly different dynamics: dengue has become hyperendemic in much of the Americas, Zika produced a single large wave in 2015–2016, and chikungunya has continued to cause spatially focused, explosive outbreaks [7, 8]. Understanding the drivers of transmission of these arboviruses at the individual, household, and community scales is a prerequisite for designing and targeting interventions [9].

Table S4: Household-size strata (households with ≥ 2 members; 163 single-person households excluded). For each size, the number of complete and incomplete households and the incomplete share.

Household size	N complete	N incomplete	% incomplete
2	133	336	71.6
3	100	540	84.4
4	77	475	86.1
5	28	214	88.4
6+	8	97	92.4

Table S5: Layer-1 (community importation) coefficients of the Longini–Koopman final-size model (main analysis, all households), on the log-odds scale and as odds ratios. Window-screen *none* (no screens) is the reference category; the intercept is the community baseline and is not exponentiated. Posterior medians with 95% credible intervals.

Parameter	log-odds (95% CrI)	OR	95% CrI
Intercept (none baseline)	−0.547 (−0.779, −0.317)	—	—
Screens: all (vs none)	−0.734 (−0.936, −0.534)	0.48	[0.39, 0.59]
Screens: some (vs none)	−0.587 (−0.781, −0.390)	0.56	[0.46, 0.68]
Local prevalence (per 10%)	0.079 (0.039, 0.119)	1.08	[1.04, 1.13]
Income: \$10–19k (vs <\$10k)	−0.083 (−0.301, 0.126)	0.92	[0.74, 1.13]
Income: $\geq$ \$20k (vs <\$10k)	−0.374 (−0.572, −0.180)	0.69	[0.56, 0.84]
AC: yes (vs no)	−0.445 (−0.612, −0.281)	0.64	[0.54, 0.76]

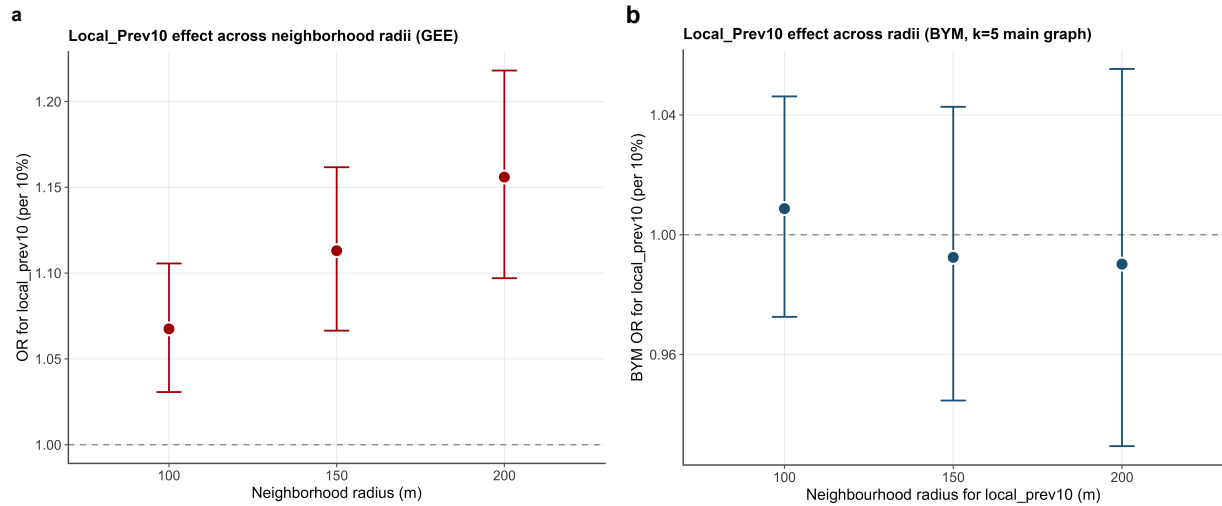


Figure S1: Local-prevalence effect across the neighbourhood radius used to define local seroprevalence (100, 150, 200 m). (a) Odds ratio for local prevalence under GEE at each radius. (b) Odds ratio for local prevalence under the spatial BYM2 model with the main ($k = 5$) adjacency graph at each radius.

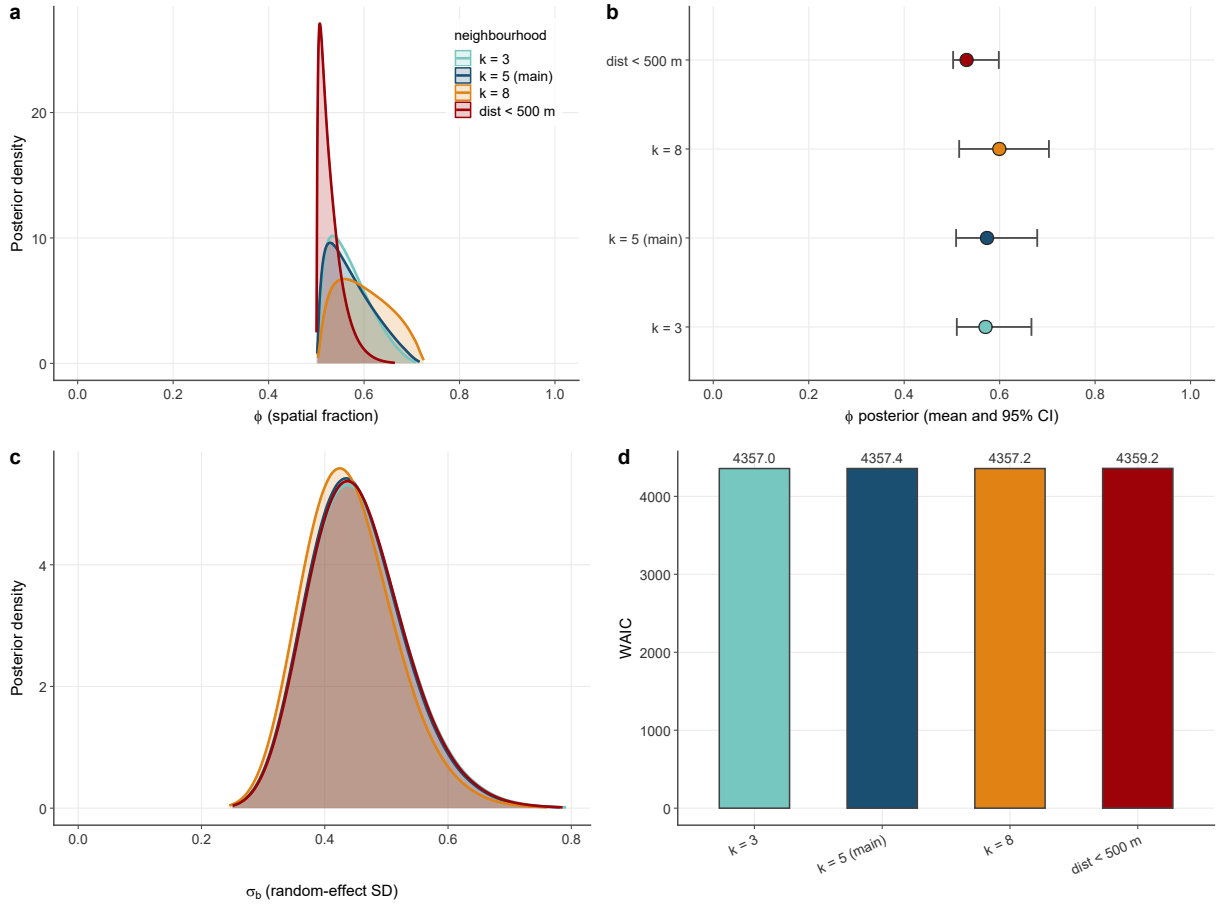


Figure S2: BYM2 spatial model across the cluster-adjacency graph ($k = 3$, $k = 5$ main, $k = 8$ nearest neighbours, and a 500 m distance threshold). (a) Posterior densities and (b) posterior means with 95% credible intervals for the structured fraction ϕ ; (c) posterior densities for the random-effect standard deviation σ_b ; (d) WAIC for each graph definition.

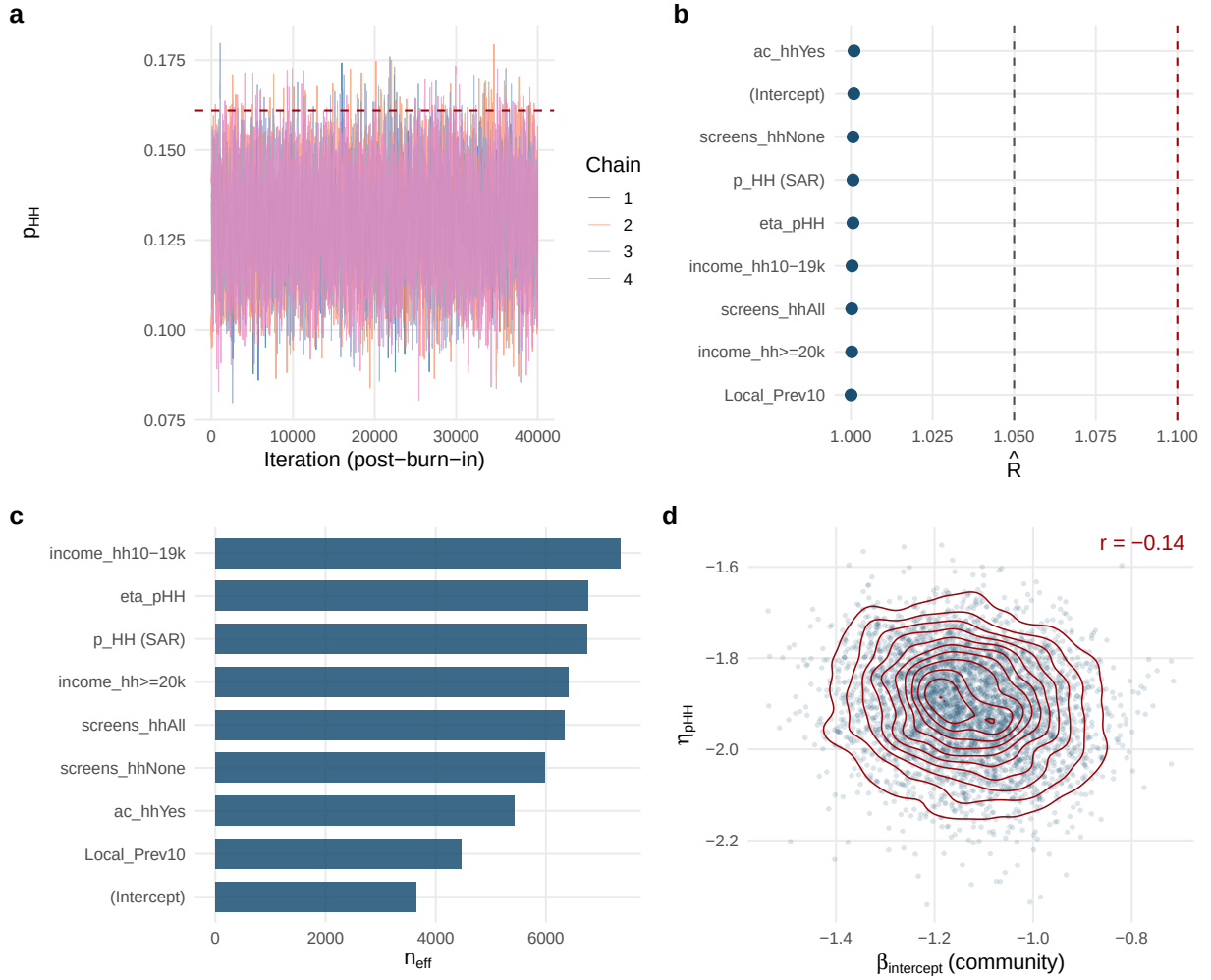


Figure S3: Convergence diagnostics for the augmented final-size MCMC. (a) Post-burn-in trace of p_{HH} across four chains; (b) Gelman–Rubin $\hat{R}$ and (c) effective sample sizes for all parameters; (d) joint posterior of the community intercept and η_{pHH} ($r = -0.14$).

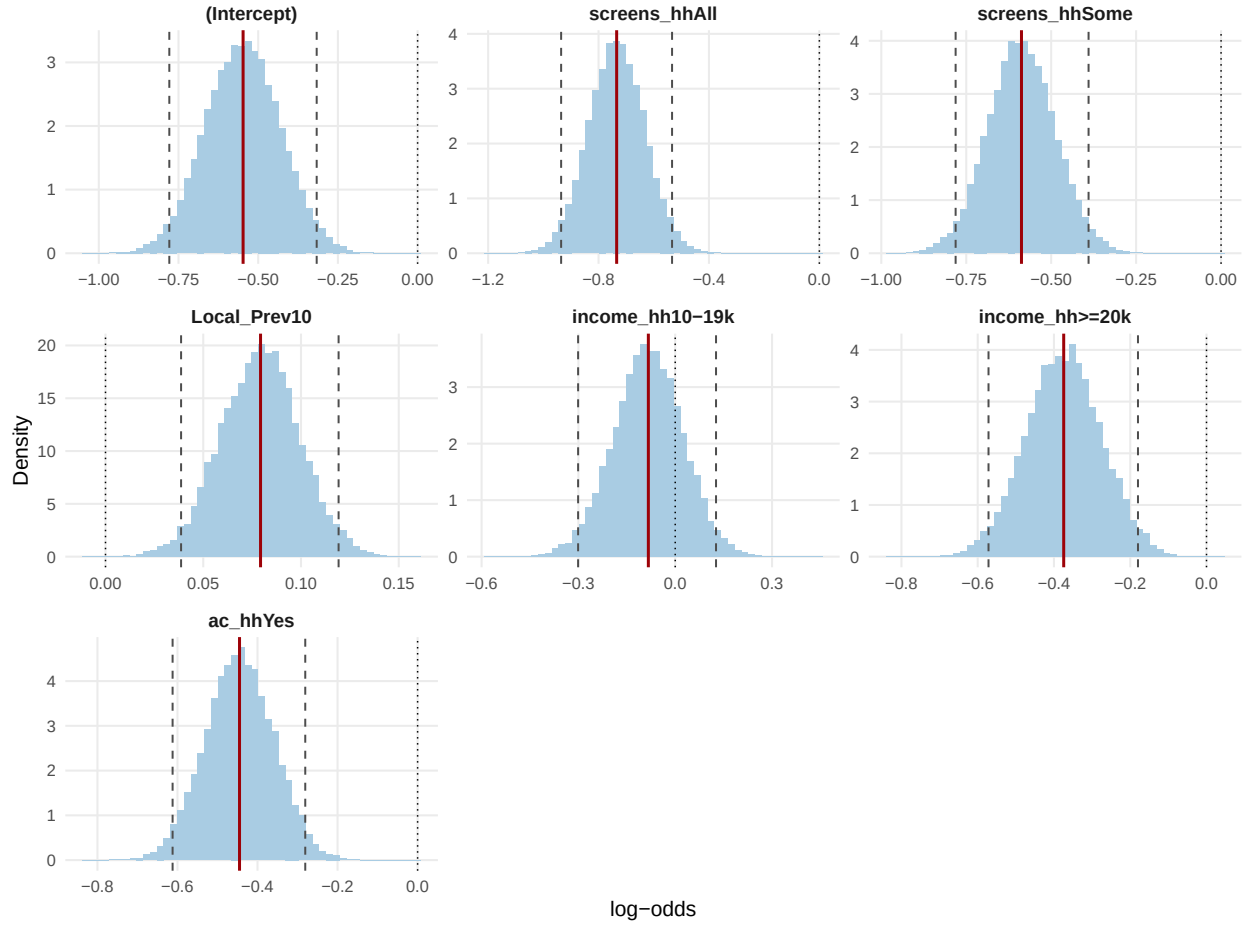


Figure S4: Marginal posterior distributions of the layer-1 (community importation) coefficients on the log-odds scale. Solid vertical lines mark posterior medians, dashed lines the bounds of the 95% credible interval, and dotted lines the null. Complete-case maximum-likelihood point estimates are overlaid.